

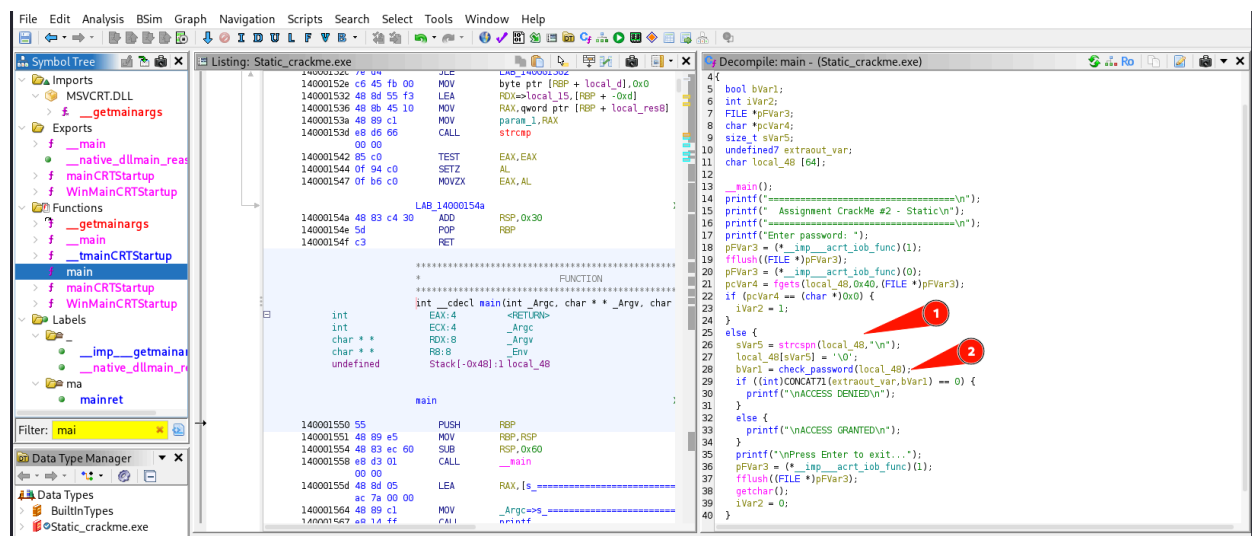
Reverse Engineering Task

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File : Static_crackme.exe

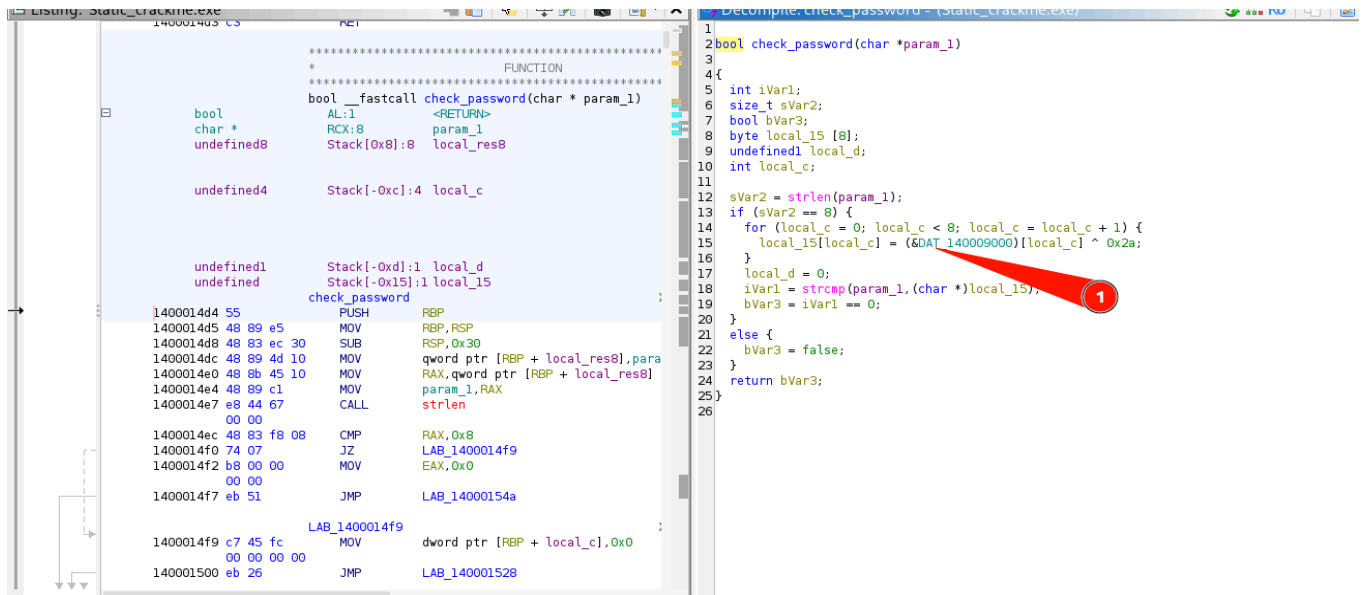
Static Analysis on Ghidra by opening the file in the ghidra UI



This program supposed to accept the user input password and store in in a temporary storage in **local_48** variable then it compares the input password and the correct hidden password then if it return 1 it prints “Access Granted” If it return 0 it prints output “ Access Dennied”

Steps to get the correct password

1 open the check_password function

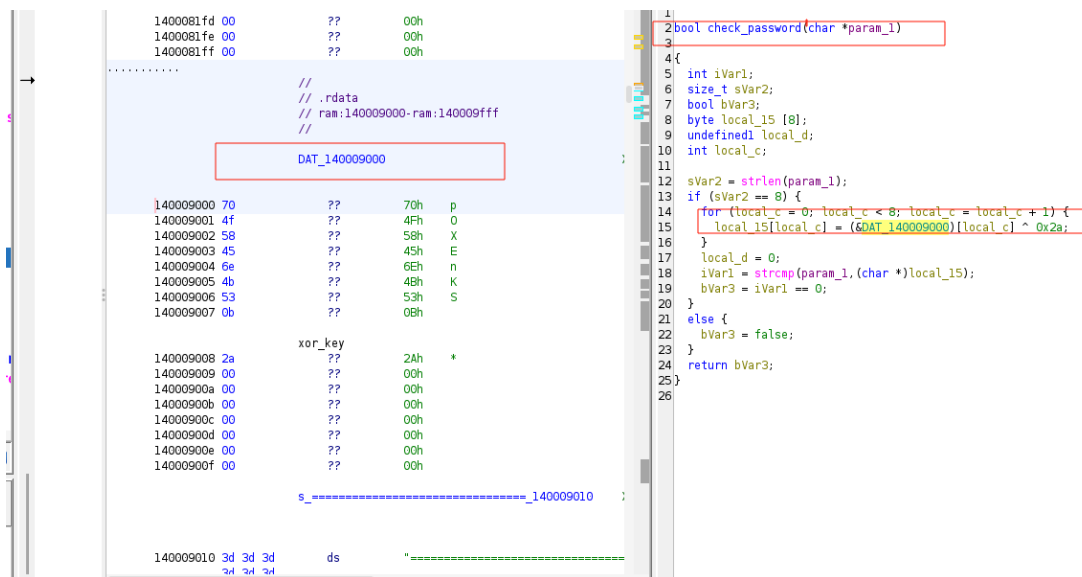


2 inside the check_password function

We observe an address where the data is stored

So we should navigate to that memory address and look for what in there

By double clicking on the “&DAT_140009000”



Observing that address its telling us the password is XORed with the key = 2a and it gives us the hexa value of the password and the key and also the plain text of the encoded password

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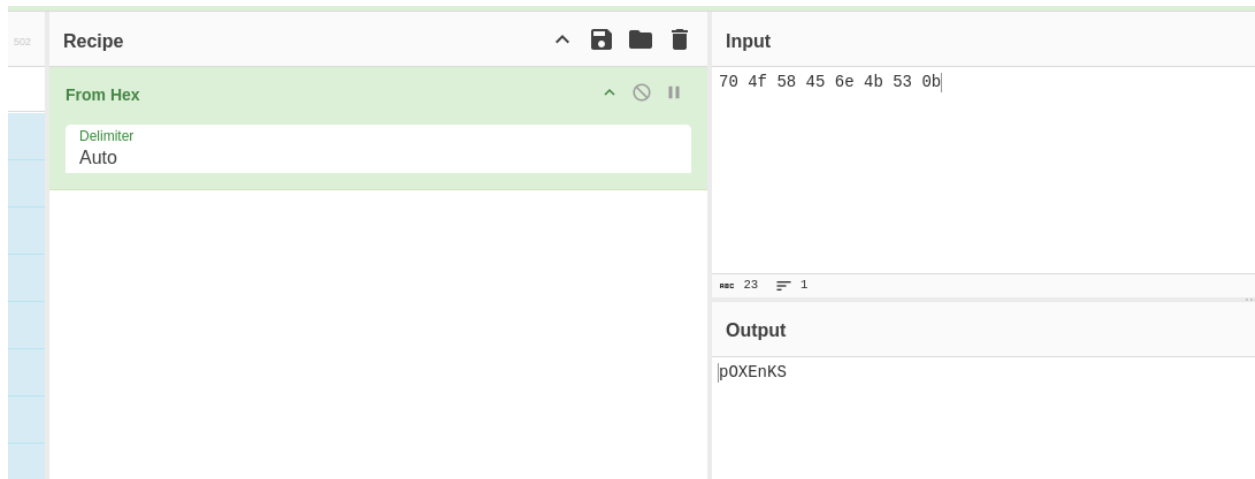
//
// .rdata
// ram:140009000-ram:140009fff
//
DAT_140009000                                XREF[3]: 1400001e4(*),
                                              check_password:140001507(*),
                                              check_password:14000150e(*)

140009000 70      ??      70h  p
140009001 4f      ??      4Fh  O
140009002 58      ??      58h  X
140009003 45      ??      45h  E
140009004 6e      ??      6Eh  n
140009005 4b      ??      4Bh  K
140009006 53      ??      53h  S
140009007 0b      ??      0Bh
xor_key
140009008 2a      ??      2Ah  *
140009009 00      ??      00h
14000900a 00      ??      00h
14000900b 00      ??      00h
14000900c 00      ??      00h
14000900d 00      ??      00h
14000900e 00      ??      00h
14000900f 00      ??      00h

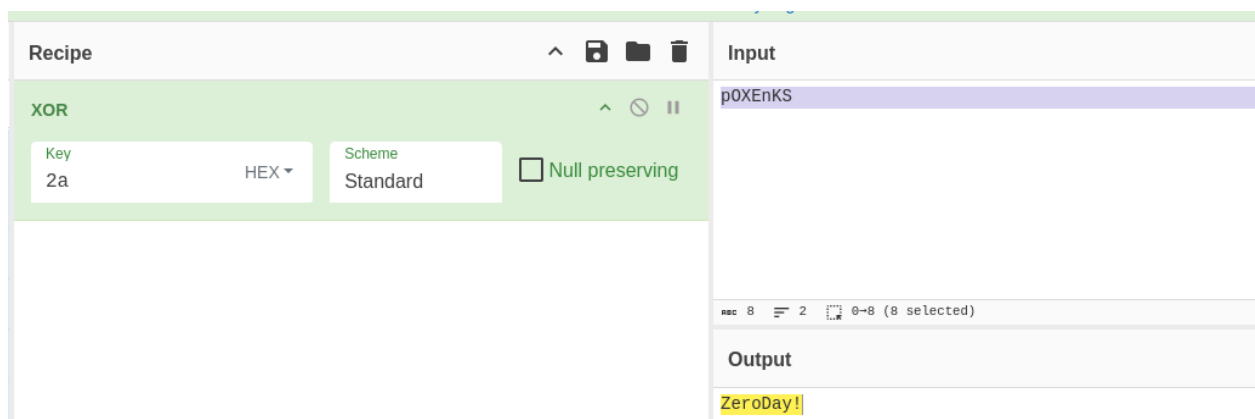
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Using CyberCheff we decode the above hexa value with kor_key=2a

First we decode that hexa value to ascii code



Final output decode that ascii to the correct password format with the xor_key=2a



The password is **ZeroDay!**

